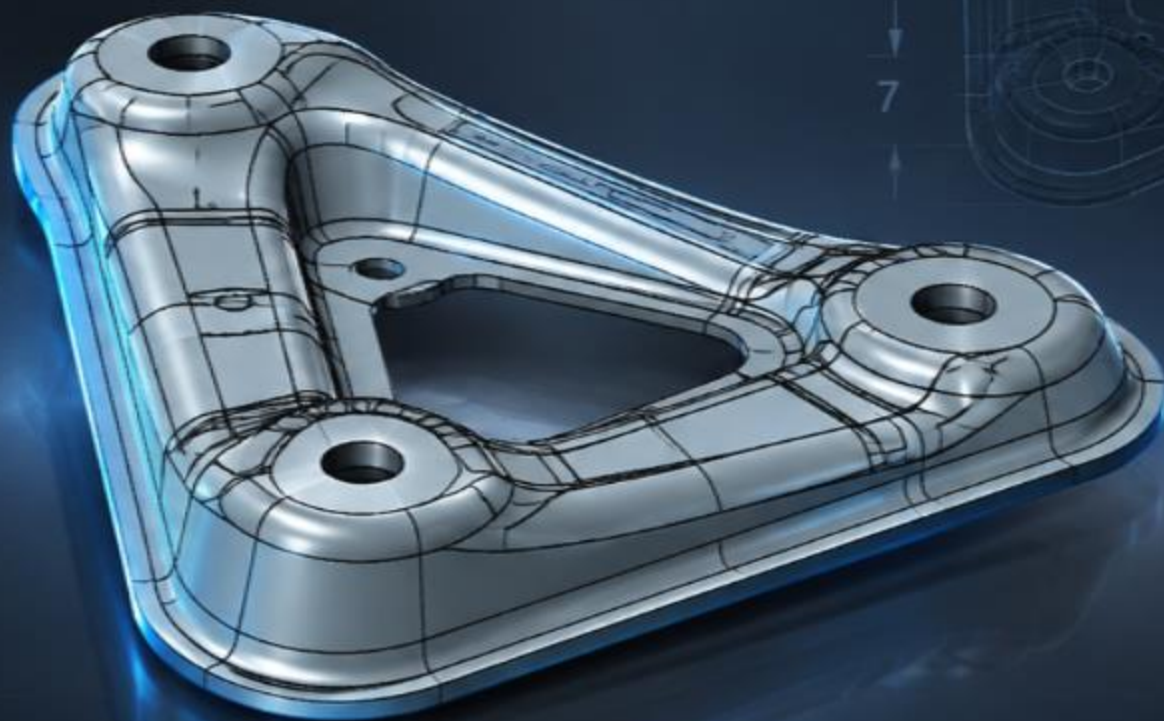




TECHBRUCE
DYNAMICS

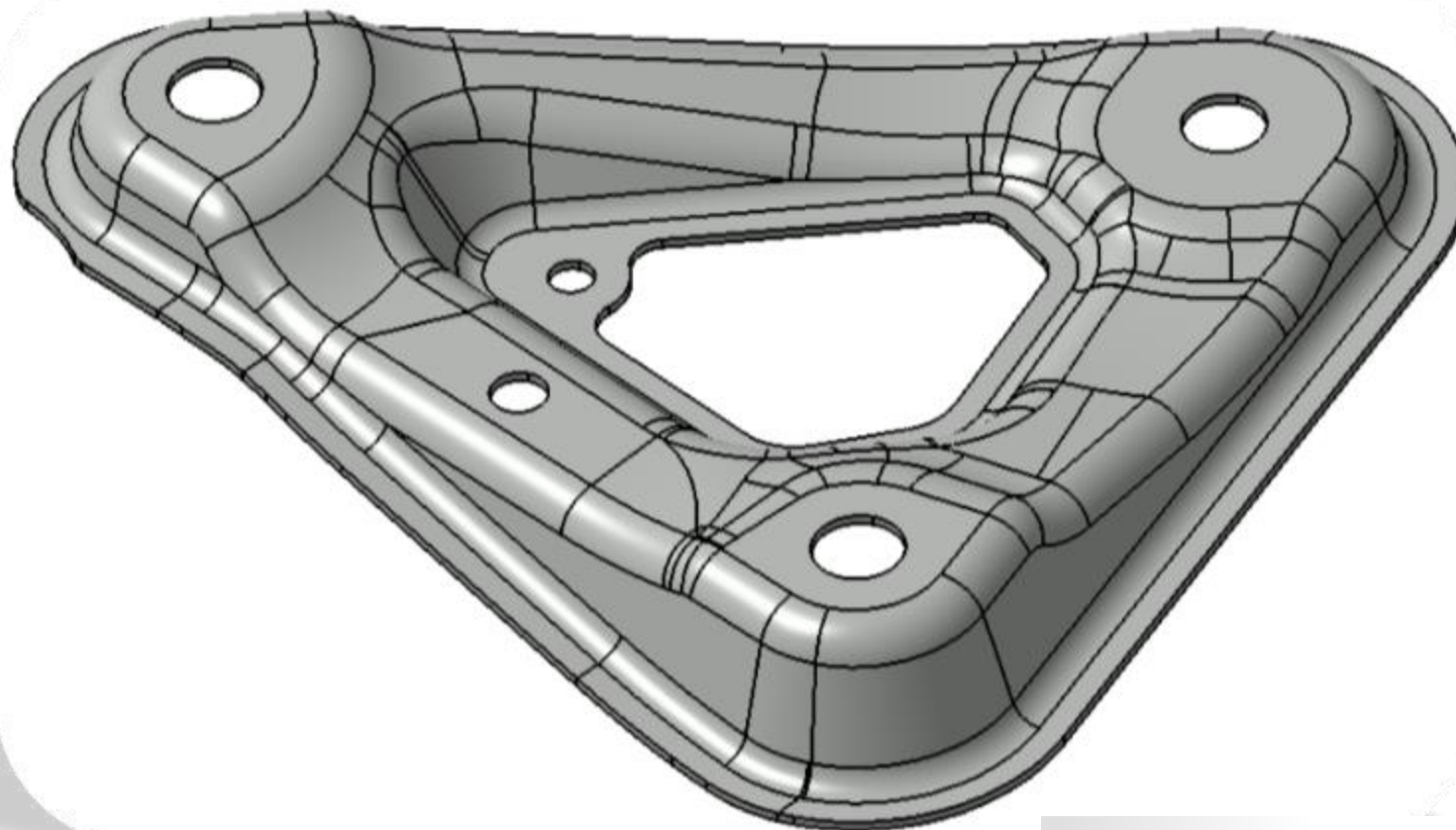
ENGINEERING IDEAS
INTO REALITY



SIMULATE | DESIGN | OPTIMIZE
FOR A BETTER TOMORROW

PRECISION
PEOPLE
POSSIBILITIES

Part Image & Details



Part Description : STAY ASSY FR SUSP MBR LH
Material : DP590
Thickness : 1.8mm

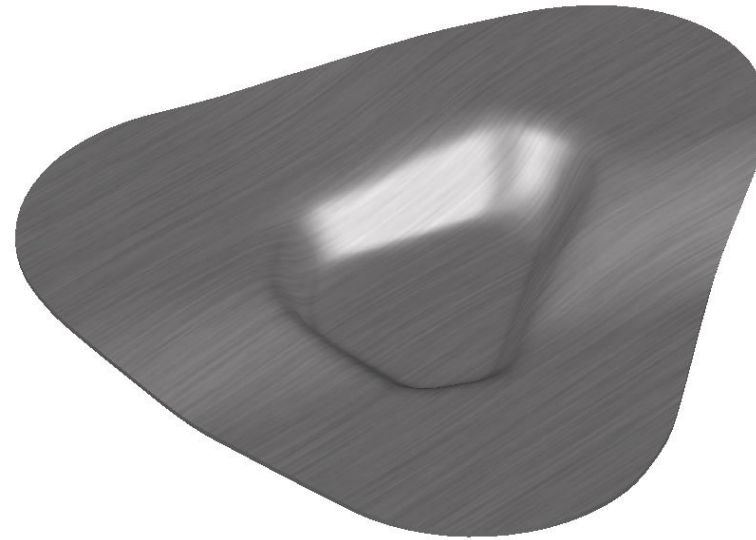
Operation sequences

Customer Proposal

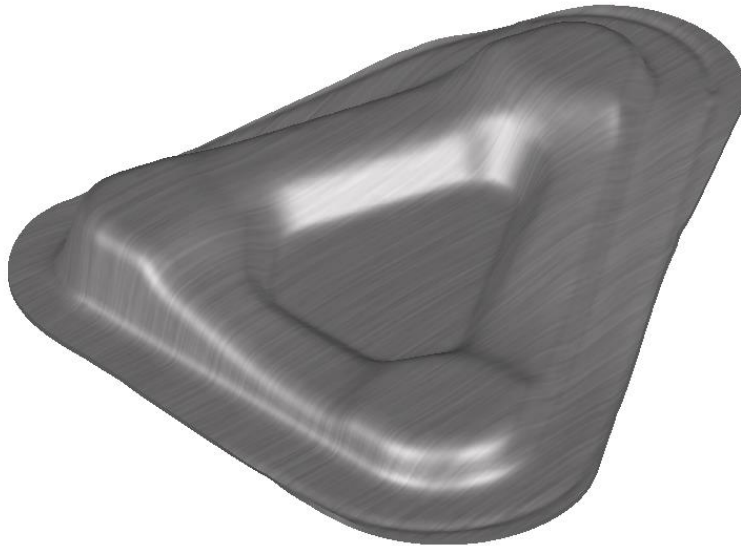
OP05-Profile Blank



OP10-Pre Draw



OP-20 Draw

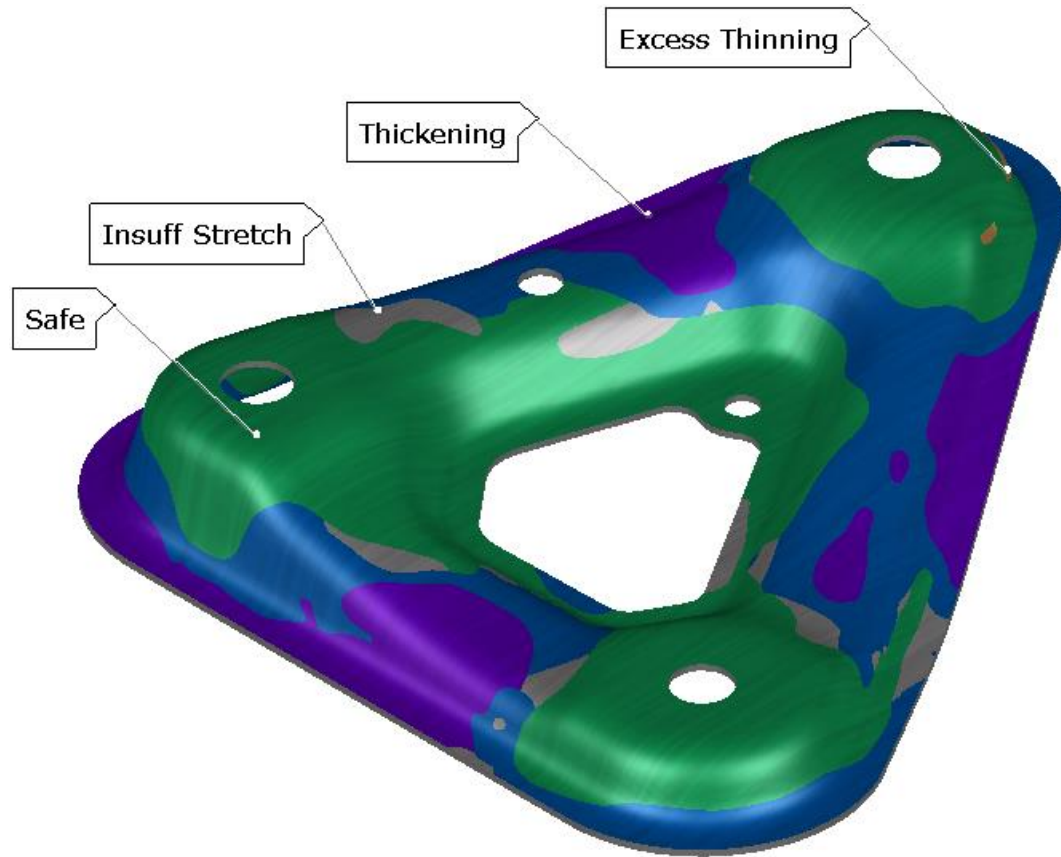


OP-20 Trim
&
Pierce

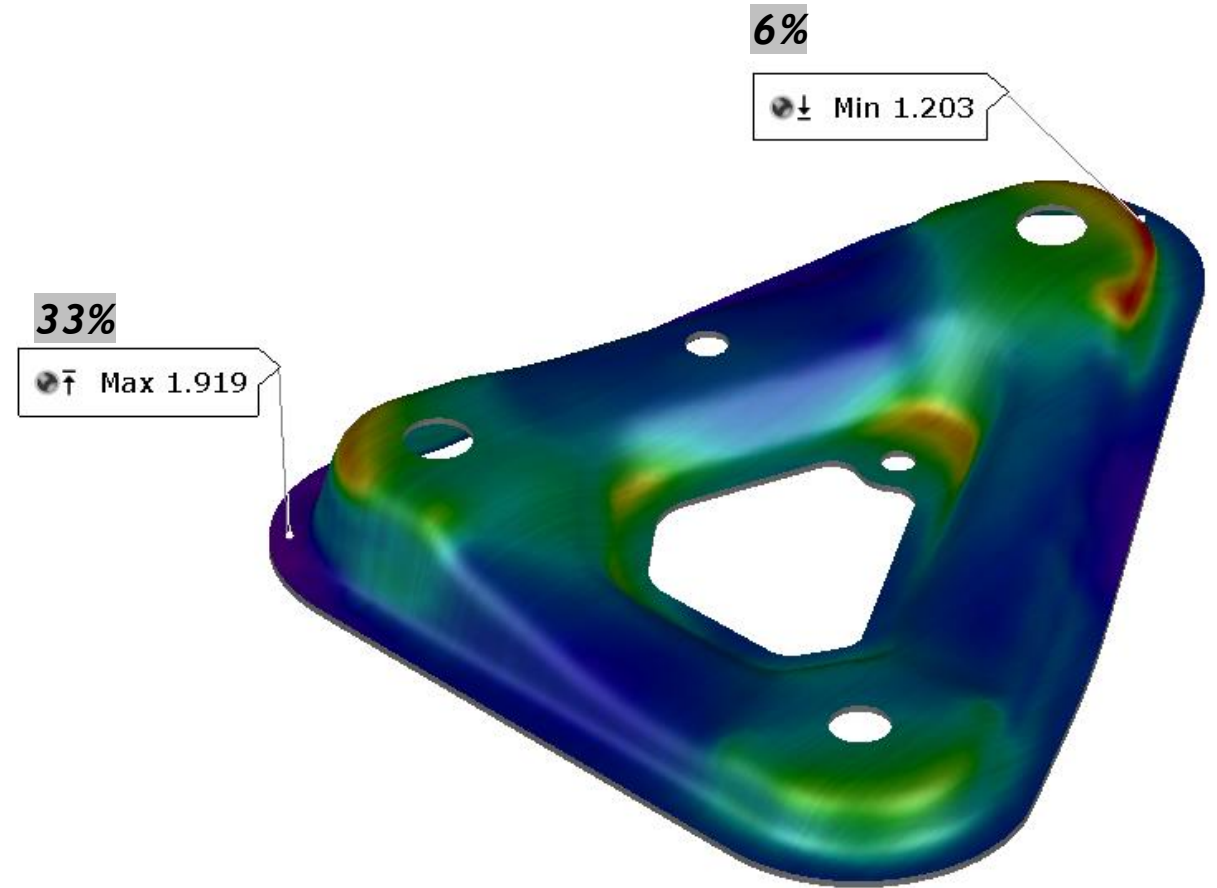


Final Part

Customer Proposal



FLD Analysis

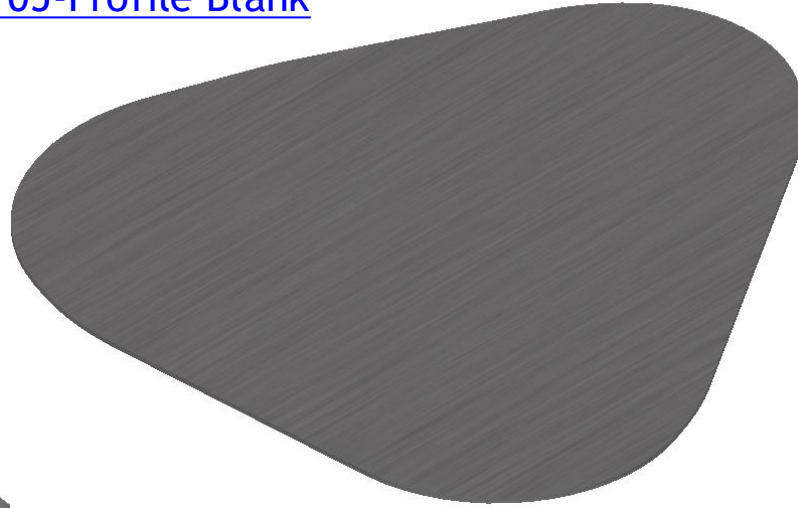


Thinning & Thickening

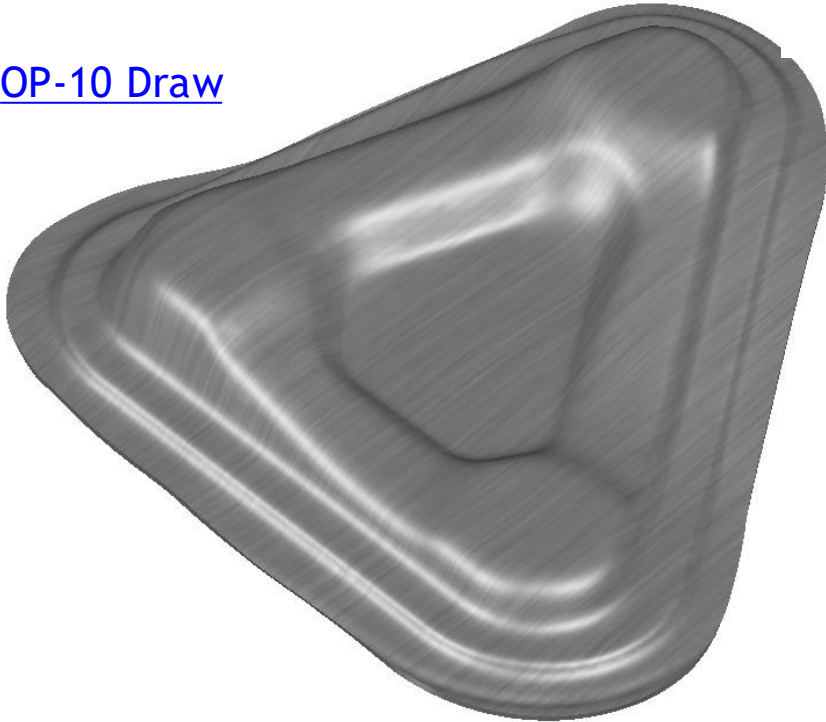
Operation sequences

Our Proposal

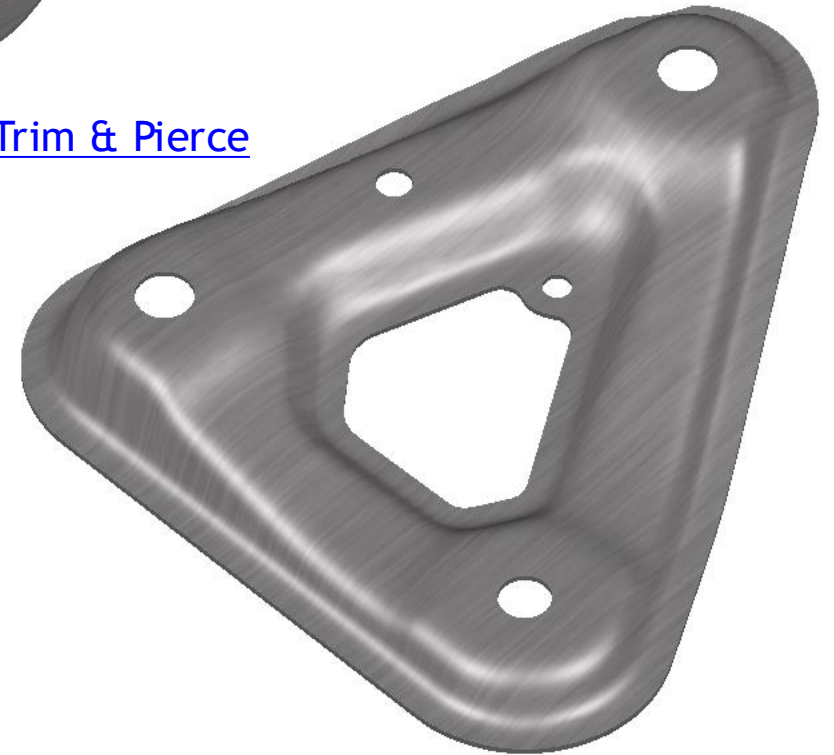
OP05-Profile Blank



OP-10 Draw

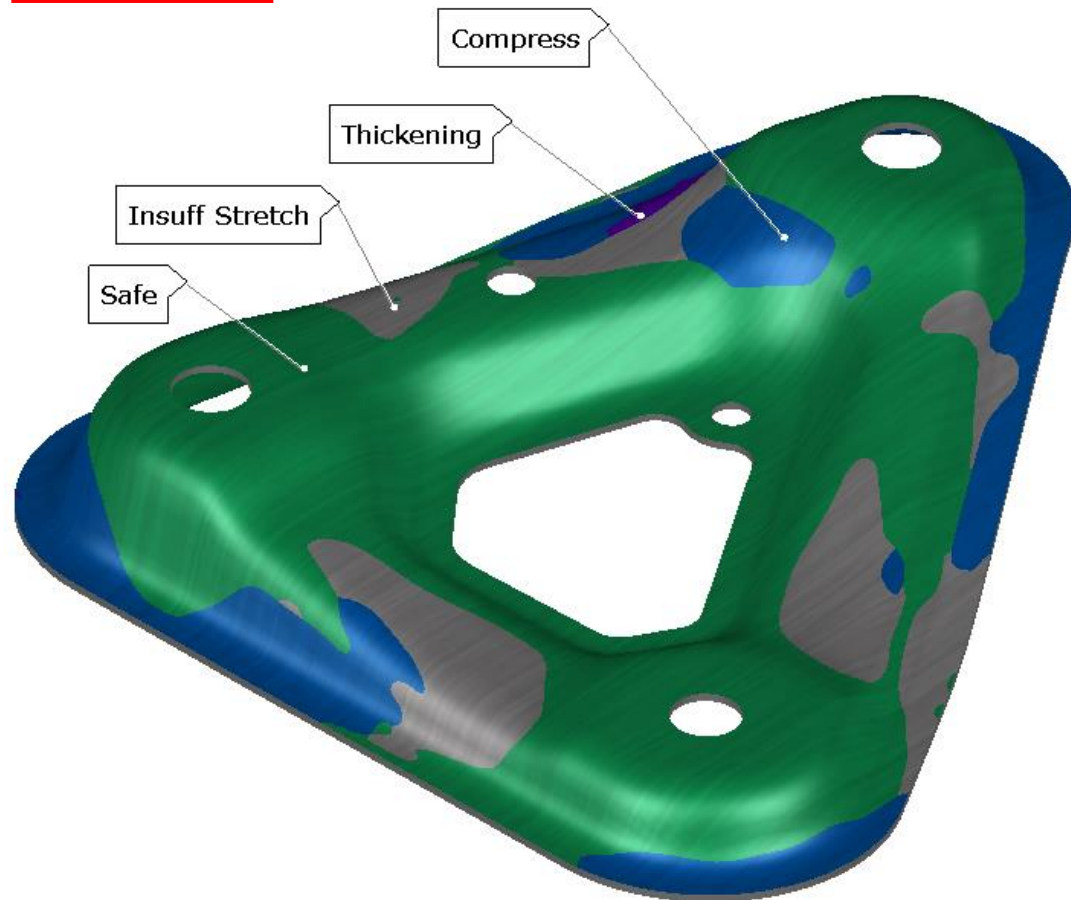


OP-20 Trim & Pierce



Final Part

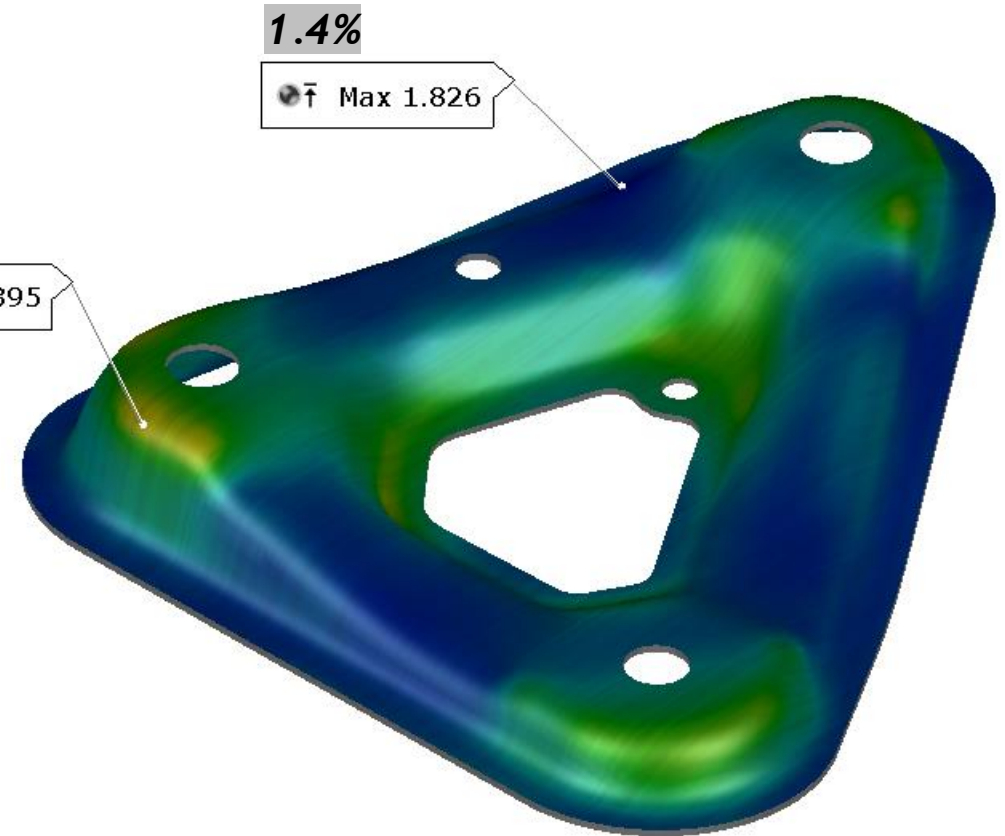
Our Proposal



FLD Analysis

22%

Min 1.395



Thinning & Thickening

Result Comparison

- ❖ **Customer:** *Profile Blank → Pre-Draw → Draw → Trim & Pierce*
- ❖ **Our Proposal:** *Profile Blank → Draw → Trim & Pierce*
- ❖ *Pre-Draw operation eliminated in our proposal.*
- ❖ *Minimum thickness improved: 1.203 → 1.395 mm*
- ❖ *Maximum thickness reduced: 1.919 → 1.826 mm*
- ❖ *More uniform thickness distribution achieved.*
- ❖ *FLD indicates acceptable formability without an additional Pre-Draw stage.*
- ❖ **Result:** *Simplified process sequence with reduced forming operations.*

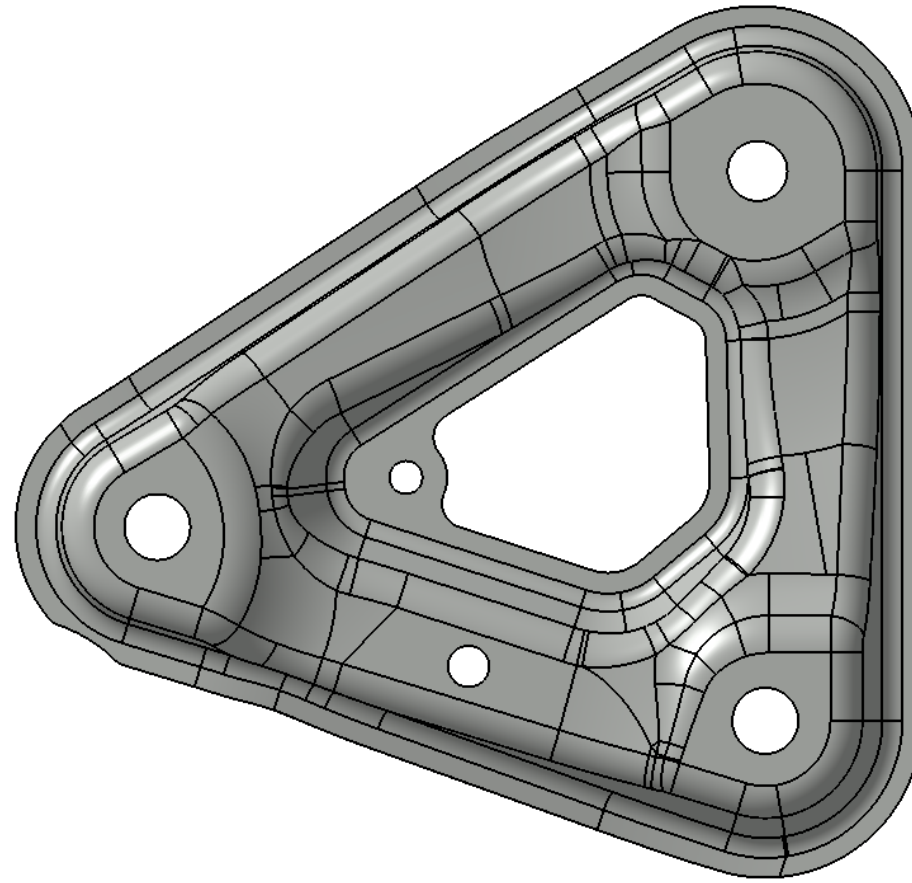
Part Geometry & Material Flow Strategy

Part Geometry:

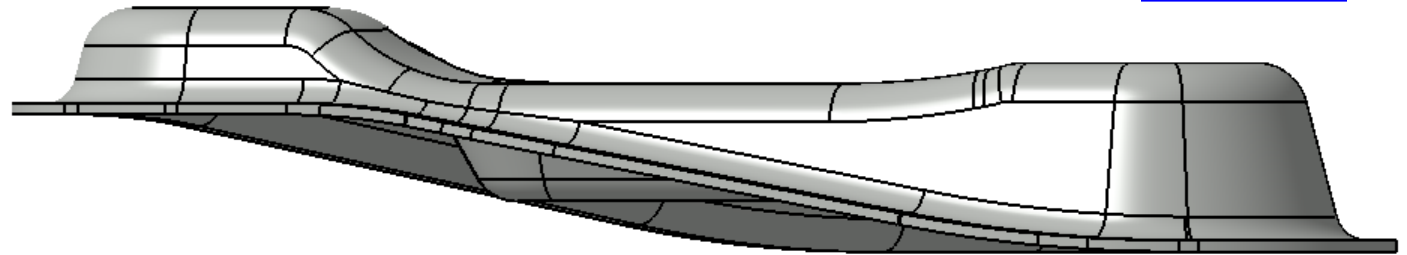
Complex geometry with varying flange depths, where the inner flange is deeper than the side flange, creating challenging material-flow conditions.

Step Usage:

A step was introduced to control material flow progressively, ensuring better material distribution and reducing excessive thinning during forming.



Top View



Front View

Draw Process Optimization

Punch & Tool Profile: The complete part profile was developed in the punch with progressive steps to distribute deformation, control material stretching, and reduce localized thinning during the draw operation.

Draw bead Layout: Positioned according to part geometry and local material-flow requirements.

Controlled Holding: Bead restraining/holding condition optimized to allow the required material flow during drawing.

Blank Optimization: Blank shape and size optimized based on material-flow behaviour.

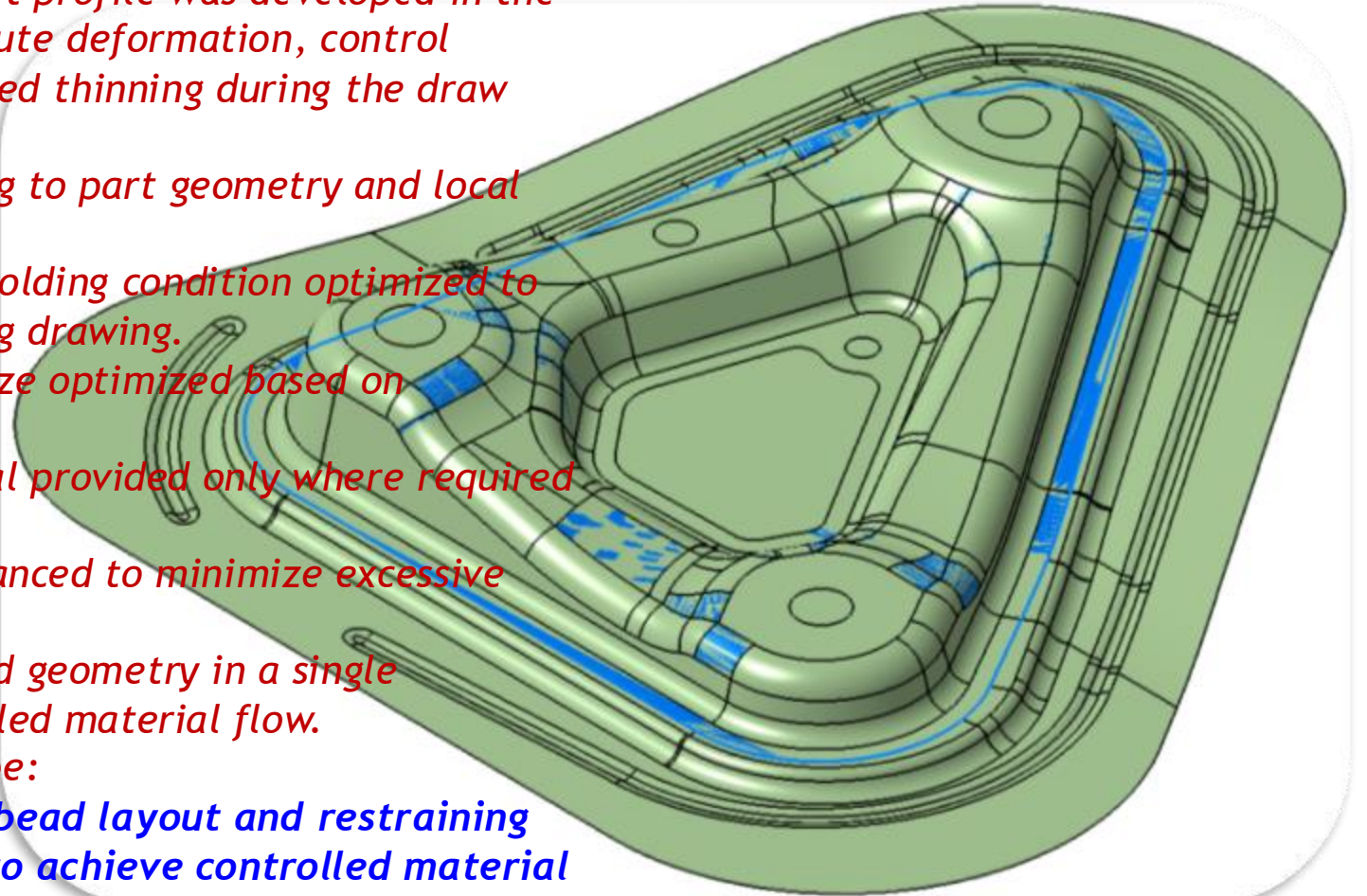
Material Utilization: Sufficient material provided only where required for forming.

Formability Control: Material flow balanced to minimize excessive thinning and thickening.

Process Objective: Achieve the required geometry in a single optimized draw operation with controlled material flow.

A strong line below the images would be:

“Part geometry, blank profile, draw bead layout and restraining conditions were optimized together to achieve controlled material flow and balanced thickness distribution.”



Our Forming Strategy

What We Do

- ❖ *Analyze material flow based on part geometry and forming severity*
- ❖ *Optimize blank profile and material availability*
- ❖ *Strategically position draw beads based on material-flow demand*
- ❖ *Control material restraining and release during forming*
- ❖ *Validate and optimize the process through simulation iterations*
- ❖ *What We Achieve*
- ❖ *Balanced and controlled material flow*
- ❖ *Sufficient material supply to critical stretch areas*
- ❖ *Reduced risk of excessive thinning and splitting*
- ❖ *Controlled thickening and wrinkling tendency*
- ❖ *Improved thickness distribution*
- ❖ *A more stable and feasible forming process*
- ❖ *Your strongest message is:*
- ❖ *“We control where the material needs to flow and where it needs to be restrained, enabling balanced stretching and improved thickness distribution.”*



Our focus is not only to validate the process, but also to identify opportunities to simplify it, reduce development effort and support a more economical tooling solution.